Department of Physics, FNSPE, CTU in Prague

02YELMA - Electricity and Magnetism

Make-up Exam

27.5.2026 / 16:00 - 17:30

Q1: A rectangular loop of dimensions a and b carries a steady current I. Write down the magnetic dipole moment $\vec{m}$ of the loop in terms of I, a, and b and find the torque $\vec{\tau}$ acting on the loop when it is placed in a uniform external magnetic field $\vec{B}$ that points at an angle θ with respect to the normal vector of the loop.

Q2: A long solenoid, of radius a, is driven by an alternating current, so that the magnetic field inside the solenoid is sinusoidal $\vec{B}(t) = B_0 \sin(\omega t)\hat{z}$ where $\hat{z}$ is the unit vector in the direction of the solenoid's axis. A circular loop of radius $a/2$ and resistance R, is placed inside the solenoid, and is coaxial with it. Find the current induced and the power dissipated in the loop as a function of time.

Q3: The magnetic field component of a monochromatic electromagnetic wave propagating in free space is given by $\vec{B}(z,t) = B_0 \sin(kz - \omega t)\hat{y}$.

(a) Find the average power transmitted through a circular area of radius a in the plane perpendicular to the direction of propagation.

(b) If the wave goes through a polarizer oriented at an angle $\pi/4$ with respect to the direction of electric field polarization, find the amplitude of the transmitted electric field.

Q4: A long cylindrical conductor has inner radius a and outer radius R ($a < R$). A steady total current I is uniformly distributed throughout the conducting material. Determine the volume current density $\vec{J}$ and find the magnetic field magnitude $B(s)$ in the regions $0 \le s < a$, $a < s < R$, and $s > R$.

Maxwell's equations

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\varepsilon_0} \qquad \nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\oint_S \vec{B} \cdot d\vec{a} = 0 \qquad \nabla \cdot \vec{B} = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt}\int_S \vec{B} \cdot d\vec{a} \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \varepsilon_0 \frac{d}{dt}\int_S \vec{E} \cdot d\vec{a} \qquad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$